



L OVELY
P ROFESSIONAL
U NIVERSITY

upGrad

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab (India)

Month: May Year: 2026

Machine Learning Project

Ola Bike Ride Request Forecast using Machine

Learning

Final Report

Submitted By

K.Vishnu Vardhan

Reddy

Reg no:

SUPERVISOR'S CERTIFICATE

This is to certify that the work reported in the B.Tech Dissertation/dissertation proposal entitled “**Ola Bike Ride Request Forecast Using Machine Learning**”, submitted by **K.Vishnu Vardhan Reddy** at **Lovely Professional University, Phagwara, India** is a bonafide record of his / her original work carried out under my supervision. This work has not been submitted elsewhere for any other degree.

Signature of Supervisor

Date:

Acknowledgement

I hereby declare that the research work reported in the dissertation/dissertation proposal entitled "Ola Bike Ride Request Forecast using ML" in partial fulfilment of the requirement for the award of Degree for Bachelor of Technology in Computer Science and Engineering at Lovely Professional University, Phagwara, Punjab is an authentic work carried out under supervision of my research supervisor **Ms. Aiswarya Saravanam**

I have not submitted this work elsewhere for any degree or diploma.

I understand that the work presented herewith is in direct compliance with Lovely Professional University's Policy on plagiarism, intellectual property rights, and highest standards of moral and ethical conduct. Therefore, to the best of my knowledge, the content of this dissertation represents authentic and honest research effort conducted, in its entirety, by me. I am fully responsible for the contents of my dissertation work.

Signature of Candidate

Name of the Candidate: K.Vishnu Vardhan Reddy

Registration No:

1. PROBLEM STATEMENT

The rapid growth of ride-sharing and bike transportation services has significantly increased the demand for efficient ride management systems. Companies such as Ola rely on accurate demand forecasting to ensure proper fleet allocation, reduce customer waiting times, and improve operational efficiency. However, predicting bike ride requests is a challenging task because demand varies depending on multiple factors such as weather conditions, temperature, humidity, wind speed, seasons, holidays, weekdays, and time periods.

The transportation industry experiences fluctuations in ride demand throughout the day and across different seasons. During favorable weather conditions and peak hours, ride requests may increase substantially, whereas adverse weather conditions, holidays, or off-peak hours may lead to lower demand. These variations make it difficult for service providers to allocate resources effectively and maintain a balance between rider demand and driver availability.

Inaccurate demand estimation can result in several operational issues. Underestimating demand may lead to insufficient bike availability, longer customer waiting times, and loss of revenue. On the other hand, overestimating demand can cause inefficient resource utilization and increased operational costs. Therefore, there is a need for a reliable and data-driven approach that can accurately forecast ride requests.

To address this challenge, this project aims to develop a Machine Learning-based Ola Bike Ride Request Prediction System. The system utilizes historical ride data and environmental factors such as season, weather, temperature, humidity, wind speed, holidays, and time-related attributes to predict the expected number of bike ride requests. By applying data preprocessing, exploratory data analysis, and predictive modeling techniques, the proposed system provides accurate ride demand forecasts and supports better decision-making for fleet management and transportation planning.

Introduction

The rapid growth of ride-sharing services has increased the need for accurate ride demand prediction. Ride requests vary based on factors such as weather conditions, temperature, humidity, seasons, holidays, and time periods. Predicting ride demand helps transportation companies optimize fleet allocation and improve customer satisfaction. This project uses Machine Learning techniques to analyze historical Ola bike ride data and forecast future ride requests. An interactive analytics dashboard is also developed to visualize trends and model performance. The system supports data-driven decision-making and enhances operational efficiency in transportation services.

Objective

The main objective of this project is to develop a Machine Learning model that can accurately predict Ola bike ride requests based on historical and environmental data. The system analyzes factors such as season, weather conditions, temperature, humidity, wind speed, holidays, and time periods to identify patterns that influence ride demand. Accurate prediction helps transportation service providers optimize fleet allocation and improve operational efficiency. The project also aims to provide an interactive analytics dashboard for visualizing ride trends and model performance. By using predictive analytics, the system supports better decision-making and enhances customer satisfaction through improved ride availability.

Scope

This project focuses on developing a Machine Learning-based system for predicting Ola bike ride requests using historical ride data. The scope includes data preprocessing, exploratory data analysis, feature engineering, model training, prediction, and performance evaluation. The dataset contains features such as season, weather conditions, temperature, humidity, wind speed, holidays, weekdays, and time periods, which are used to train the prediction models. Machine Learning algorithms such as Linear Regression, Ridge Regression, and Lasso Regression are utilized to identify patterns influencing ride demand. The project also includes the development of an interactive dashboard for data visualization and demand forecasting. However, the scope is limited to the available dataset and does not include real-time traffic data, GPS tracking, or live weather integration.

Literature Review

Several studies have explored the use of Machine Learning techniques for demand forecasting in transportation and ride-sharing services. Traditional statistical methods and Linear Regression models were initially used due to their simplicity and ease of interpretation. However, researchers found that ride demand is influenced by multiple factors such as weather conditions, temperature, humidity, seasons, holidays, and time periods, making the prediction problem more complex.

To improve prediction accuracy, advanced Machine Learning algorithms such as Ridge Regression, Lasso Regression, Decision Trees, Random Forests, and Gradient Boosting have been widely adopted. Research shows that proper data preprocessing, feature scaling, and handling missing values significantly enhance model performance. Many transportation and ride-sharing platforms use predictive analytics to optimize fleet management and improve customer satisfaction. This project builds upon these findings by applying regression-based machine learning models to accurately predict Ola bike ride requests and provide valuable insights through an interactive analytics dashboard.

Dataset Selection & Preprocessing

DATASET SELECTION:

Dataset Description

- Number of rows and columns: **10,886 rows, 13 columns.**
- Feature Types:

Categorical:

- Season
- Weather
- Weekday
- AM/PM
- Holidays

Numerical:

- Temperature
- Humidity
- Windspeed
- Casual Riders
- Count (Ride Requests)
- Year
- Month
- Day

The dataset contains historical bike ride booking information along with environmental and time-related factors, making it suitable for building a reliable ride demand prediction model.

Column Descriptions

- **season:** Represents the season during which the ride request was recorded (Spring, Summer, Fall, or Winter).
- **weather:** Describes the weather condition at the time of the ride request, such as clear, cloudy, rainy, or snowy.
- **temp:** Records the temperature during the ride request period.
- **humidity:** Represents the moisture content present in the atmosphere.
- **windspeed:** Indicates the wind speed measured during the ride request period.
- **casual:** Represents the number of casual or non-registered riders using the bike service.
- **count:** Total number of bike ride requests recorded during a specific period. This is the target variable used for prediction.

- **year:** Indicates the year in which the ride request was recorded.
- **month:** Represents the month of the year.
- **day:** Represents the day of the month.
- **weekday:** Indicates whether the ride request occurred on a weekday or a weekend.
- **am_or_pm:** Specifies whether the ride request was recorded during the AM or PM time period.
- **holidays:** Indicates whether the day was a holiday or a regular working day.

Data Cleaning & Feature engineering

Rows with missing values:

```
df.isnull().sum()
```

```
datetime      0
season        0
weather       0
temp        1632
humidity      1632
windspeed     1632
casual        0
registered    0
count        0
dtype: int64
```

Strategy for handling missing values:

- The dataset was examined for missing values using Pandas data analysis techniques..
- Missing values were found in the numerical features **temp**, **humidity**, and **windspeed**, with 1,632 missing records in each column.
- Since these features are important for predicting bike ride demand, the missing values were replaced using the **median value** of their respective columns.
- Median imputation was selected because it is less affected by extreme values and preserves the overall distribution of the data.
- No missing values were found in the remaining features such as season, weather, casual riders, registered riders, and ride count.
- After handling the missing values, the dataset was converted into a clean and structured format suitable for machine learning model training and prediction.

DATA STANDARDIZATION AND ENCODING

DATA STANDARDIZATION AND ENCODING

1. Standardization (Z-score Scaling) was applied to numerical features:

- temp
- humidity
- windspeed
- casual
- registered

Standardization transforms these features by subtracting the mean and dividing by the standard deviation, ensuring that they have a mean of 0 and a standard deviation of 1. This helps prevent machine learning models from being biased toward larger numerical values and improves training stability.

2. Label Encoding was used for categorical features:

- season
- weather
- **Spring = 1** The categorical values were converted into numerical form so that machine learning algorithms could process them effectively. For example:
 - Summer = 2
 - Fall = 3
 - Winter = 4

Similarly, weather conditions were encoded into numerical values representing different weather categories.

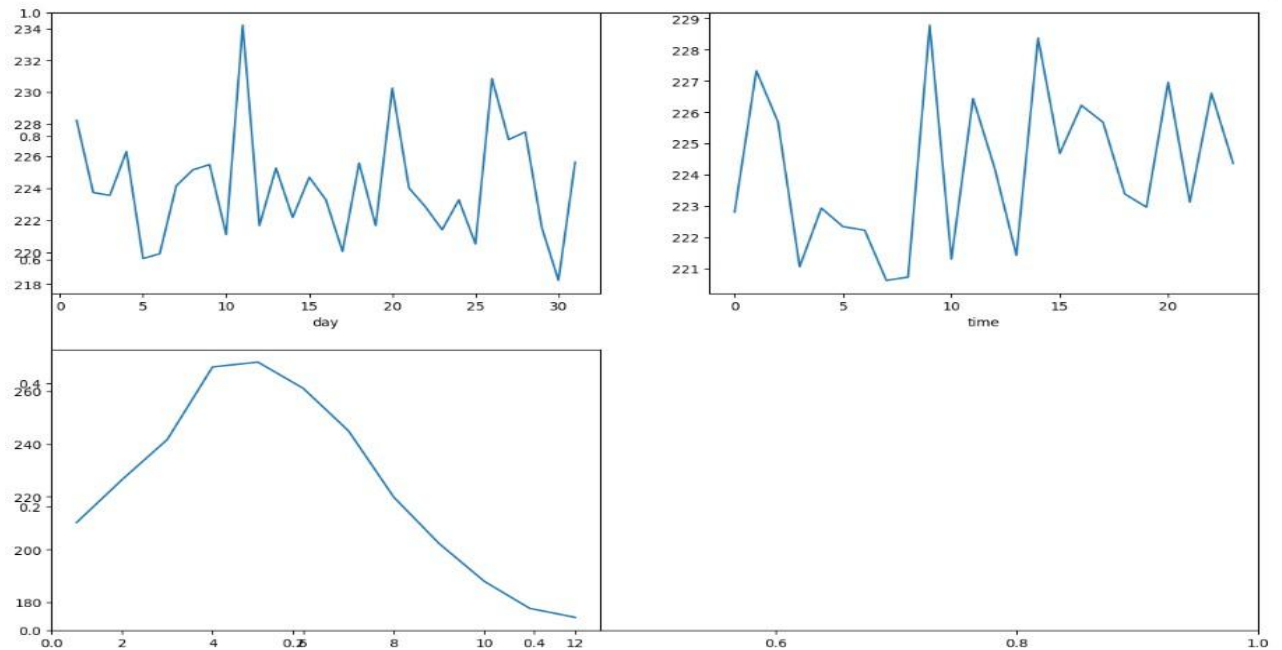
3. Feature Engineering

Additional features such as time-related variables and environmental conditions were analyzed to improve prediction performance. Factors including season, weather, temperature, humidity, wind speed, and rider information were used to capture ride demand patterns more effectively.

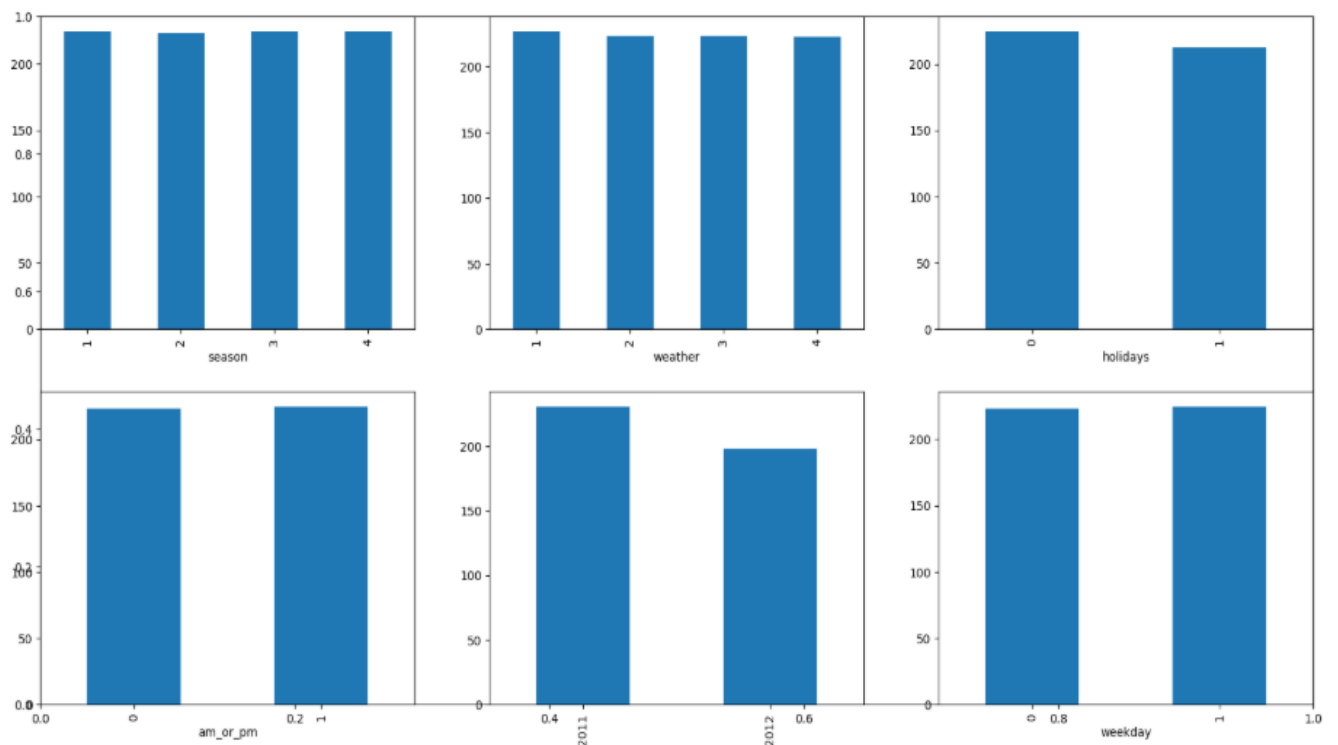
By applying these preprocessing techniques, the dataset was transformed into a format that ensures fair comparisons across features, improved model efficiency, and better predictive accuracy.

Exploratory Data Analysis

```
features = ['day', 'time', 'month']
plt.subplots(figsize=(15, 10))
for i, col in enumerate(features):
    plt.subplot(2, 2, i + 1)
    df.groupby(col).mean()['count'].plot()
plt.show()
```



```
features = ['season', 'weather', 'holidays',\
            'am_or_pm', 'year', 'weekday']
plt.subplots(figsize=(20, 10))
for i, col in enumerate(features):
    plt.subplot(2, 3, i + 1)
    df.groupby(col).mean()['count'].plot.bar()
plt.show()
```



Machine Learning Algorithms

Linear Regression

Linear Regression is one of the most widely used machine learning algorithms for predicting continuous numerical values. It establishes a linear relationship between independent variables and the target variable. In this project, Linear Regression was used to predict the number of bike ride requests based on factors such as season, weather conditions, temperature, humidity, wind speed, holidays, and time-related attributes. The model is simple, easy to interpret, and serves as a strong baseline for evaluating prediction performance.

Ridge Regression

Ridge Regression is an extension of Linear Regression that includes a regularization term to reduce overfitting. It penalizes large coefficient values and improves the model's ability to generalize on unseen data. In this project, Ridge Regression helped improve prediction stability by handling multicollinearity among input features and reducing model complexity while maintaining good predictive performance.

Lasso Regression

Lasso Regression is another regularized version of Linear Regression that performs both prediction and feature selection. It automatically reduces the coefficients of less important features to zero, thereby selecting only the most relevant variables. In this project, Lasso Regression helped identify the most influential factors affecting ride demand while reducing unnecessary complexity in the model.

Model Training

The dataset was divided into training and testing sets using an 80:20 ratio. Approximately 80% of the data was used for training the machine learning models, while the remaining 20% was reserved for testing. Three regression models—Linear Regression, Ridge Regression, and Lasso Regression—were trained and compared. The models learned the relationship between environmental factors, time-related variables, and ride demand patterns. Their performance was then evaluated using regression metrics on unseen test data.

Evaluation Metrics

Since this is a regression problem involving the prediction of ride request counts, the following metrics were used:

- **R² Score (Coefficient of Determination):** Measures how well the model explains the variation in ride demand.

- **Mean Squared Error (MSE):** Calculates the average squared difference between actual and predicted values.
- **Root Mean Squared Error (RMSE):** Represents the average prediction error in an interpretable form by taking the square root of MSE.

Model Performance

The performance of Linear Regression, Ridge Regression, and Lasso Regression models was evaluated using R^2 Score, Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). These metrics were used to measure the prediction accuracy and error of each model on the test dataset. The comparative evaluation helped assess the effectiveness of the regression models in predicting bike ride demand. The model with the highest R^2 Score and the lowest MSE and RMSE values was considered the best-performing model. This performance analysis helped in understanding the prediction capability and generalization ability of each model on unseen data.

Comparative Analysis

In this project, three regression models were compared: Linear Regression, Ridge Regression, and Lasso Regression. All three models were used to predict bike ride requests based on environmental and time-related factors such as season, weather conditions, temperature, humidity, wind speed, holidays, and time periods.

Linear Regression served as the baseline model and established a direct relationship between the input features and ride demand. It provided good prediction results and was easy to interpret. However, its performance was affected when features showed correlations with one another.

Ridge Regression improved upon Linear Regression by introducing regularization. This technique reduces the impact of large coefficient values and helps prevent overfitting. As a result, Ridge Regression produced more stable predictions and better generalization on unseen data.

Lasso Regression also uses regularization but has the additional advantage of performing feature selection. It automatically reduces the coefficients of less important features to zero, allowing the model to focus on the most significant variables influencing ride demand. This helps simplify the model while maintaining prediction accuracy.

The comparison of these models was carried out using evaluation metrics such as R^2 Score, Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). The model with the highest R^2 Score and the lowest error values was considered the best-performing model. The results demonstrated that regularized regression techniques, particularly Ridge and Lasso Regression, can improve prediction performance and provide more reliable forecasts for bike ride demand.

Overall, the comparative analysis showed that machine learning models can effectively capture ride demand patterns and support data-driven decision-making for transportation services.

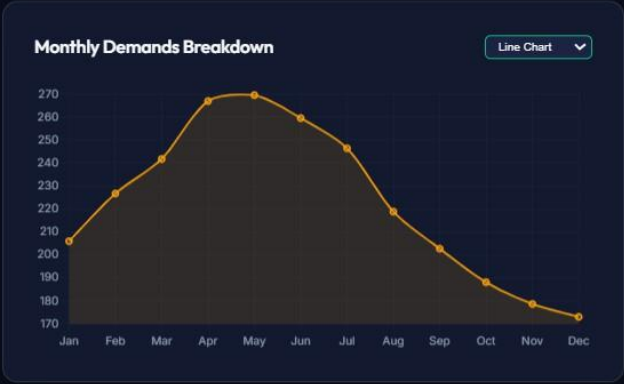
Project Deployment

Dashboard

Ride Analytics Dashboard

Observe the distribution of ride request counts based on temporal and environmental features.

Export Analytics Report



Conclusion

This project successfully developed a Machine Learning-based system for predicting Ola bike ride requests using historical ride data and environmental factors. Various features such as season, weather conditions, temperature, humidity, wind speed, holidays, weekdays, and time periods were analyzed to understand their impact on ride demand.

The dataset was preprocessed, explored, and used to train multiple regression models including Linear Regression, Ridge Regression, and Lasso Regression. The models were evaluated using regression metrics such as R^2 Score, Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) to measure prediction accuracy and performance.

The results demonstrated that machine learning techniques can effectively forecast ride demand and help transportation service providers make informed decisions regarding fleet allocation and resource management. The interactive analytics dashboard further enhanced the project by providing meaningful visualizations and insights into ride demand patterns.

Overall, the project highlights the importance of predictive analytics in modern transportation systems and shows how machine learning can improve operational efficiency, reduce waiting times, and support data-driven decision-making. Future enhancements may include real-time weather integration, traffic analysis, and advanced machine learning models for even more accurate demand forecasting.

References

1. Kaggle Dataset – Ola Bike Ride Request Dataset.
2. Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
3. McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*.
4. Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array Programming with NumPy. *Nature*, 585, 357–362.
5. Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering*, 9(3), 90–95.
6. Python Software Foundation. Python Programming Language Documentation. <https://www.python.org>
7. Scikit-learn Documentation. <https://scikit-learn.org>
8. Pandas Documentation. <https://pandas.pydata.org>
9. Matplotlib Documentation. <https://matplotlib.org>
10. Chart.js Documentation. <https://www.chartjs.org>